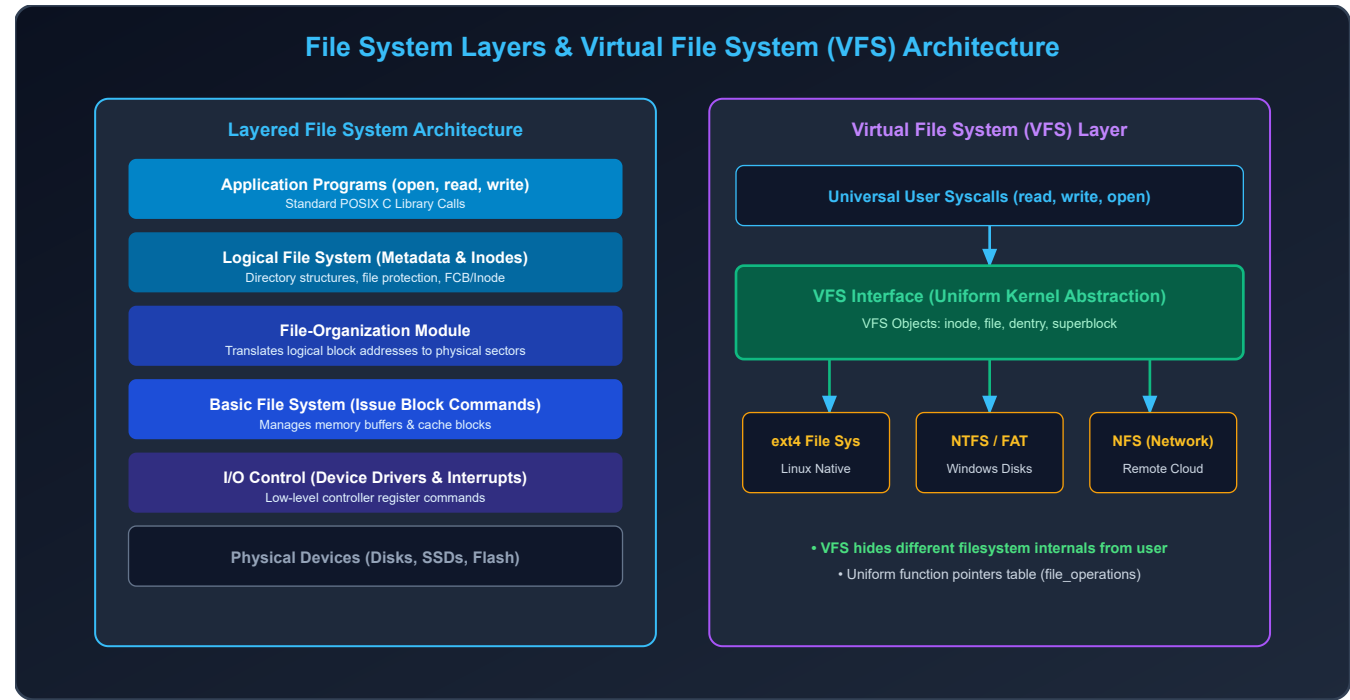


Module 09: File System Implementation and VFS

1. On-Disk vs In-Memory Structures

Structure	Location	Stored Information
Superblock	Disk Partition	Total blocks, free block count, inode list
Inode / FCB	Disk & Cached in RAM	File size, owner, permissions, block pointers
System-Wide Open Table	RAM	Global open files, reference count
Per-Process Open Table	RAM (PCB)	File descriptor (fd), current seek offset



2. Virtual File System (VFS)

VFS application programs ko uniform POSIX API deta hai jabki internally ext4, NTFS, FAT, aur NFS ke driver implementations ko dynamically dispatch karta hai.